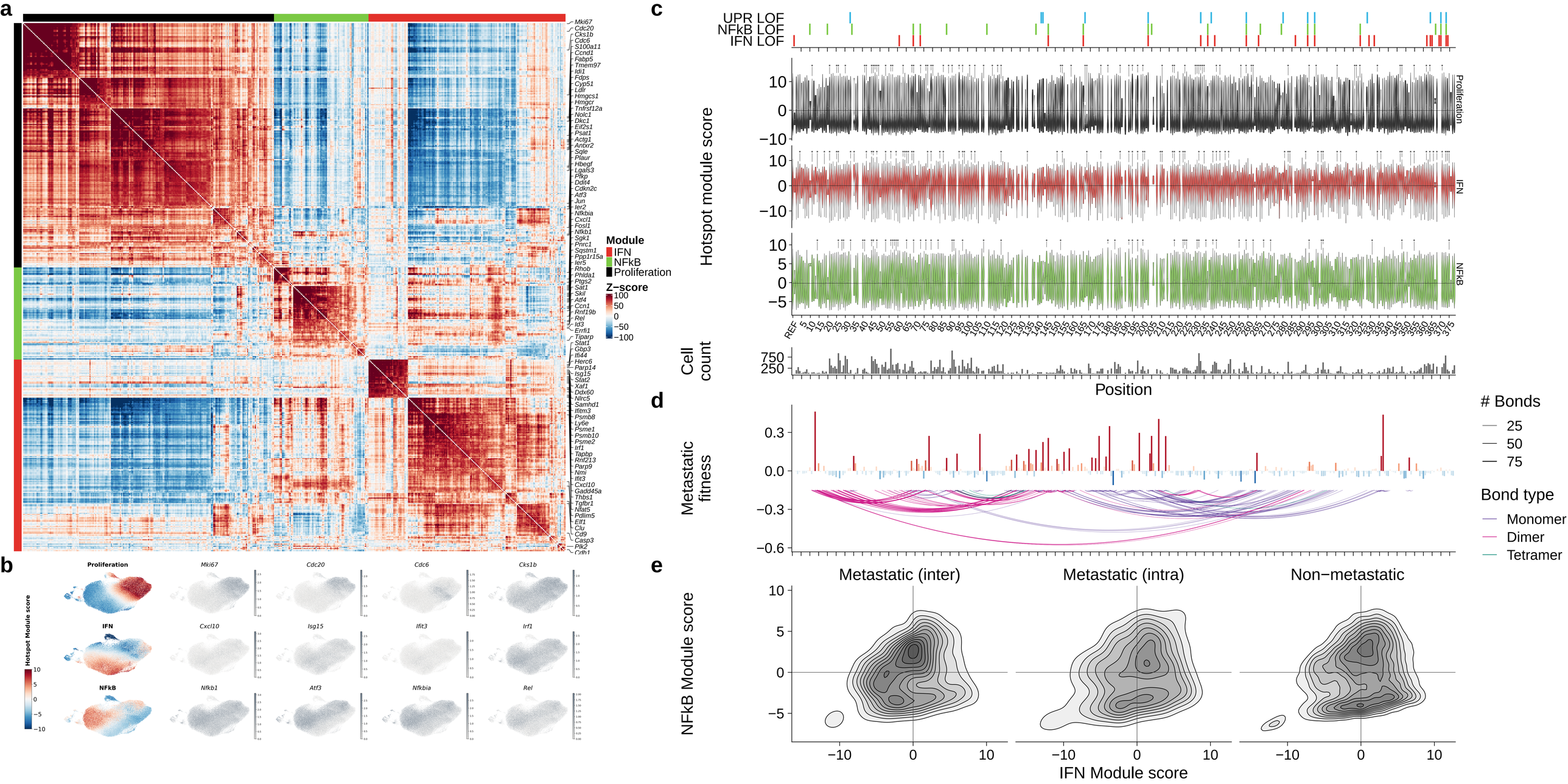

a
Module
IFN
NFkB
Proliferation
Z−score
100
50
0
−50
−100
b
Proliferation
IFN
NFkB
Hotspot Module score
c
UPR LOF
NFkB LOF
IFN LOF
Hotspot module score
Proliferation
IFN
NFkB
Cell count
Position
d
Metastatic fitness
Bonds
25
50
75
Bond type
Monomer
Dimer
Tetramer
e
NFkB Module score
Metastatic (inter)
Metastatic (intra)
Non−metastatic
IFN Module score